

16. The Cost of Lumber, continued The analysis of variance F -test in Exercise 15 (Section 11.2) indicated that there was a difference in the average cost of lumber for the four states. The following information from that exercise is given in the table:

Sample Means	$\bar{x}_1 = 262.2$	MSE	41.25
	$\bar{x}_2 = 234.8$	Error df:	16
	$\bar{x}_3 = 251.6$	n_i :	5
	$\bar{x}_4 = 268.6$	k :	4

Use Tukey's method for paired comparisons to determine which means differ significantly from the others at the $\alpha = .01$ level.

$$W = q_{\alpha, k, df} \cdot \frac{\sqrt{MSE}}{\sqrt{n_i}}$$

$$= q_{0.01}(4, 16) \cdot \frac{\sqrt{41.25}}{\sqrt{5}}$$

$$= 5.19 \cdot \frac{\sqrt{41.25}}{\sqrt{5}} \approx 14.91$$

2.892

$$\#. |\bar{x}_i - \bar{x}_j| > W \Rightarrow \text{significant difference}$$

$$|\bar{x}_1 - \bar{x}_2| = 27.4 > 14.91 \Rightarrow \text{significant difference}$$

$$|\bar{x}_1 - \bar{x}_3| = 10.6 < 14.91 \Rightarrow \text{no significant difference}$$

$$|\bar{x}_1 - \bar{x}_4| = 6.4 < 14.91 \Rightarrow \text{no significant difference}$$

$$|\bar{x}_2 - \bar{x}_3| = 16.8 > 14.91 \Rightarrow \text{significant difference}$$

$$|\bar{x}_2 - \bar{x}_4| = 33.8 > 14.91 \Rightarrow =$$

$$|\bar{x}_3 - \bar{x}_4| = 17.2 > 14.91 \Rightarrow =$$